u_Pointwise_Conv

Pointwise_Conv

u_SystolicArray

SystolicArray

clken
load_mode
rst

psum_out[255:0]

w_in[127:0]

x_in[127:0]

ROWS[0].COLS[0].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[0].COLS[1].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[0].COLS[2].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[0].COLS[3].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[0].COLS[4].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[0].COLS[5].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[0].COLS[6].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[0].COLS[7].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

ROWS[1].COLS[0].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[1].COLS[1].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[1].COLS[2].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[1].COLS[3].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[1].COLS[4].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[1].COLS[5].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[1].COLS[6].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[1].COLS[7].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

ROWS[2].COLS[0].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[2].COLS[1].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[2].COLS[2].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[2].COLS[3].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[2].COLS[4].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[2].COLS[5].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[2].COLS[6].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[2].COLS[7].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

ROWS[3].COLS[0].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[3].COLS[1].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[3].COLS[2].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[3].COLS[3].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[3].COLS[4].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[3].COLS[5].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[3].COLS[6].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[3].COLS[7].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

ROWS[4].COLS[0].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[4].COLS[1].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[4].COLS[2].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[4].COLS[3].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[4].COLS[4].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[4].COLS[5].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[4].COLS[6].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[4].COLS[7].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

ROWS[5].COLS[0].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[5].COLS[1].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[5].COLS[2].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[5].COLS[3].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[5].COLS[4].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[5].COLS[5].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0] ROWS[5].COLS[6].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[5].COLS[7].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

ROWS[6].COLS[0].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[6].COLS[1].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[6].COLS[2].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[6].COLS[3].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[6].COLS[4].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[6].COLS[5].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[6].COLS[6].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]

x_in[15:0]

x_out[15:0]

ROWS[6].COLS[7].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] w_out[15:0]
x_in[15:0] ROWS[7].COLS[0].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] x_in[15:0]

x_out[15:0]

ROWS[7].COLS[1].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] x_in[15:0]

x_out[15:0]

ROWS[7].COLS[2].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] x_in[15:0]

x_out[15:0]

ROWS[7].COLS[3].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] x_in[15:0]

x_out[15:0]

ROWS[7].COLS[4].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] x_in[15:0]

x_out[15:0]

ROWS[7].COLS[5].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] x_in[15:0]

x_out[15:0]

ROWS[7].COLS[6].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] x_in[15:0]

x_out[15:0]

ROWS[7].COLS[7].pe_inst

ProcessingElement

clken
load_mode

rst psum_in[31:0] psum_out[31:0] w_in[15:0] x_in[15:0]